

The double-edged sword of endorsing external ideas: Juggling competitive advantage and organizational compatibility concerns

Rebekah SungEun Hong^a, Vijaya Venkataramani^b, Mengxi Yang^{c,*}

^a Department of Management, Gatton College of Business and Economics, University of Kentucky, 550 South Limestone, Lexington, KY 40506-0034, United States

^b Department of Management and Organization, Robert H. Smith School of Business, University of Maryland, College Park, 4544 Van Munching Hall, College Park, MD 20742, United States

^c School of Economics and Management | MOE Social Science Laboratory of Digital Economic Forecasts and Policy Simulation, University of Chinese Academy of Sciences, Beijing 100190, PR China

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ABSTRACT

In spurring employee innovation, organizations often encourage employees to seek creative ideas from external sources. However, research findings on managers' receptivity to external ideas are mixed. While some work suggests that managers are favorable towards external ideas, other studies indicate that they often exhibit a "not-invented-here" syndrome, a negative attitude towards external ideas. Drawing from Dynamic Capabilities Theory and integrating it with the managerial creativity endorsement literature, we develop a dual-pathway model that argues that there are competing considerations managers face when evaluating external creative ideas. Using a field experiment as well as multiple lab studies, we show that managers consider both the risk of being out-competed by other organizations if they do *not* adopt these ideas, as well as the risk of incompatibility with existing systems within their organization if they *do* implement them. While the former makes managers more receptive to external ideas, the latter risk makes them averse to implementing them. We further demonstrate managers' perceptions of technological turbulence in the environment as a crucial boundary condition that amplifies the tension between these risks. We discuss the theoretical and practical implications of our findings.

Employee innovation is a critical means by which organizations differentiate themselves from competitors (Shalley, Zhou, & Oldham, 2004; West, 1990). An important catalyst for such innovation is ideas and inspiration from diverse external sources that can provide fresh insights that are different from what is available internally in the organization (Laursen & Salter, 2006; Lichtenthaler, 2011; Song, Almeida, & Wu, 2003). However, as some research indicates, organizations face challenges in adopting external ideas due to the "not-invented-here" (NIH) syndrome, a common bias where decision-makers view externally sourced knowledge, such as ideas or technologies, with skepticism (Antons & Piller, 2015; Katz & Allen, 1982; Lucas & Goh, 2009). For example, despite the growing success of cloud services from Amazon (AWS) and Google, Microsoft was slow to shift its focus from its traditional on-premises software business to the cloud. Yet, there is substantial evidence that some firms also actively seek, replicate, or adapt external knowledge and practices (Haunschild, 1993; Menon & Pfeffer, 2003; Piezunka & Dahlander, 2015). For instance, Amazon's one-click purchase feature inspired other online retailers to adopt similar

systems to avoid losing market share (Gauri et al., 2021). What explains these divergent reactions to innovative external ideas? What factors impact this inherent tension between embracing external creativity and battling internal skepticism?

In addressing these important questions, we uniquely integrate insights from Dynamic Capabilities Theory (DCT; Teece, 2007) with the literature on managerial creativity endorsement (e.g., Mueller et al., 2012; Zhou et al., 2017). DCT research, primarily formulated at the firm level, highlights the importance of external ideas as a key source of the knowledge and innovation necessary for competitive advantage. DCT argues that firms need to sense and react to opportunities and threats in the external environment while also ensuring the smooth reconfiguration of internal processes when deciding to adopt innovations. These strategic decisions to implement new ideas emerge not only from the actions of top management but also from those of numerous lower-level employees and their managers (Mintzberg & Waters, 1985; Reitzig & Sorenson, 2013). Yet, psychological research on managerial creativity endorsement finds that managers tend to prematurely reject novel ideas

* Corresponding author.

E-mail addresses: Rebekah.hong@uky.edu (R.S. Hong), vvenkata@umd.edu (V. Venkataramani), yangmengxi@ucas.ac.cn (M. Yang).

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from employees because they perceive these ideas to be uncertain, and thus risky (e.g., Johnson & Proudfoot, 2024; Lane et al., 2022).

Integrating these two literatures, we bring attention to the *competing* risk considerations underlying managers' decisions to implement ideas originally developed externally. Because managers are held accountable for their decisions, they are motivated to strengthen the firm's competitive position while also avoiding decisions that could lead to failure (Levitt & March 1988; March & Shapira, 1987). This results in the need to balance two types of risk: the risk of omission (failing to act, which could mean losing an opportunity to competitors) and the risk of commission (acting and potentially causing harm; e.g., Kahneman & Tversky, 1982; Ritov & Baron, 1990). Echoing this dual imperative, DCT (e.g., Teece et al., 1997) emphasizes the need to seize opportunities, yet manage the risk of reconfiguring existing systems in implementing innovation. Extending this framework, we argue that managers weigh two competing risks when considering external ideas: the risk of being outpaced by competitors if they do not adopt such ideas (risk of omission; e.g., Abrahamson, 1991) and the risk of incompatibility with current systems if they do (risk of commission; e.g., Chan, Li, & Zhu, 2018; Nelson & Winter, 1982). We propose that while the perceived risk of losing competitive ground encourages managers to endorse external ideas, the risk of incompatibility leads them to hesitate in adopting them.

In examining when the tension between these competing risk considerations might be exacerbated, we draw further guidance from DCT. DCT suggests that a firm's ability to reconfigure resources, processes, and routines, as well as capture new opportunities and address emerging threats, becomes especially important in rapidly changing technological environments (Teece et al., 1997). When managers perceive high technological turbulence—characterized by rapid advancements and frequent shifts in technology—they experience increasing pressure to stay current to maintain a competitive edge. Unfortunately, however, this can also generate much uncertainty about new ideas' compatibility with existing processes and systems. Ironically, this means that in situations where external ideas are especially critical (Chen et al., 2018; Ciborra, 1996), managers' perceptions of such environments can heighten the tensions already underlying their decisions about such ideas. Fig. 1 illustrates our model.

Our paper makes some important contributions. First, by integrating the micro literature on managerial creativity endorsement with more macro research on innovation management (i.e., Dynamic Capabilities Theory), it provides a nuanced treatment of how managers decide to implement external creative ideas. Although the psychological literature on idea endorsement has focused on why managers are unwilling to implement creative ideas in general (e.g., Mount et al., 2021; Mueller et al., 2012; Venkataramani et al., 2023, 2024), it has not systematically investigated this in the context of externally developed ideas. By applying insights from DCT to highlight the competing factors

underlying decisions to endorse external versus internal ideas, we aim to shed more light on this issue.

Second and related, it unpacks the risk-based considerations that underlie managers' endorsement of external creative ideas. The managerial creative idea endorsement research (albeit not focused on external ideas) has focused on broad conceptions of risk to explain why managers reject novel ideas in general. Specifically, while this past work has primarily treated risk as a deterrent (Johnson & Proudfoot, 2024; Lane et al., 2022), our research demonstrates the more nuanced effects of different types of risk. Understanding the tension between these risks is important because it allows managers to assess trade-offs between adopting external ideas and maintaining stability, and to choose the right time to implement external innovations instead of adopting reactive decision-making and disruptive changes all at once.

Finally, while most managerial creativity endorsement research has focused on the characteristics of the idea, target, receiver, or organizational context (e.g., Lu et al., 2019; Mueller et al., 2018; Zhou et al., 2019), our work emphasizes how managers' perceptions of a key environmental factor—technological turbulence—act as an important boundary condition in affecting their different risk assessments and willingness to implement external ideas. Our findings reveal the irony that high levels of perceived turbulence increase the tensions associated with leveraging external creativity, especially when such external ideas are most needed.

1. Theory and hypotheses

Employee innovation consists of creativity—the generation of novel and useful ideas—as well as its implementation (Anderson, Potocnik, & Zhou, 2014). To spur innovation, organizations often provide employees with opportunities to attend conferences and networking events to gain exposure to new ideas (Afuah & Tucci, 2012; Laursen & Salter, 2006). Yet, when employees do propose innovative ideas, many of these ideas do not get implemented (Mueller et al., 2012). One of the main bottlenecks explaining this “innovation gap” has been middle managers (e.g., Mueller et al., 2012; Lu et al., 2019; Zhou et al., 2017). This is because although mandates for innovation usually come from top management, they are often enacted by middle-level managers, who are the gatekeepers that evaluate employee ideas and make decisions about resource allocation and implementation (Mollick, 2012; Reitzig & Sorenson, 2013). Unfortunately, however, recent work suggests that these managers often reject or are unwilling to implement creative ideas because they are averse to the uncertainty that the novelty of these ideas entails (Mueller et al., 2012). Given the uncertainty surrounding novel ideas, there is also more variability in judgments of the value of such ideas. Decision-makers interpret this as a signal of risk, thus reducing their willingness to invest in the idea (Johnson & Proudfoot, 2024; Lane et al., 2022). In addition, various manager individual differences (Mount et al.,

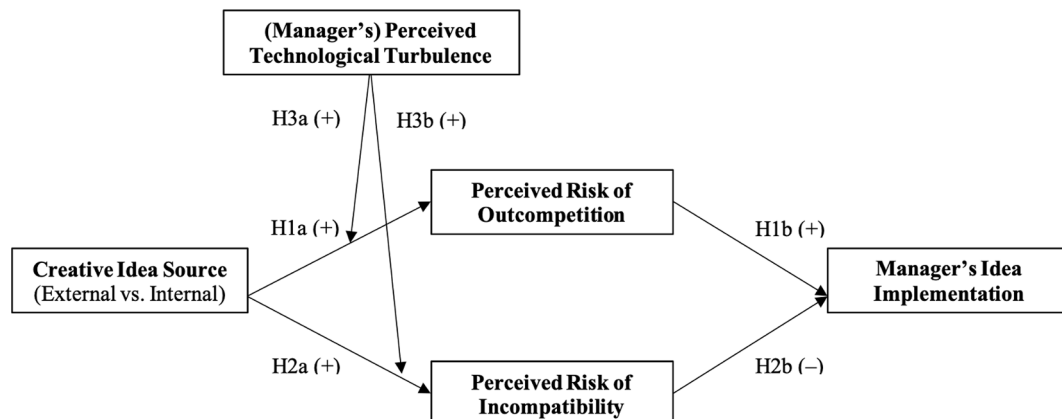


Fig. 1. Conceptual Model.

2021; Sijbom et al., 2016; Venkataramani et al., 2023), as well as contextual factors (see Zhou et al., 2017 for a review) have been shown to affect managers' rejection of creative ideas.

While most of this managerial creativity endorsement research has examined creative ideas without differentiating between those developed externally versus internally within organizations, other studies highlight an idea's source as a key factor influencing decision-makers. For example, Criscuolo et al. (2017) found that selection panels tended to reject innovative ideas from applicants who were not from their location. Importantly, research on the "Not-Invented-Here" (NIH) syndrome, though not directly undertaken in the context of creativity, has documented a strong negative bias toward external ideas (Antons & Piller, 2015). For instance, Reitzig and Sorenson (2013) documented an in-group bias where managers favored ideas from individuals within their own division while undervaluing ideas from outside sources. Organizations frequently perceive in-house knowledge creation as more legitimate and prestigious than external innovations (Burcharth et al., 2014; Michailova & Husted, 2003) and resist external ideas as they may threaten pride in local expertise or raise fears of perceived incompetence (Agrawal et al., 2010; Lüttgens et al., 2014; see also Haslam et al., 2013).

Although this has been the dominant narrative, there is also competing evidence that firms seek ideas generated externally, as such ideas provide new insights that are not available internally (Haunschild, 1993). For example, organizations frequently use crowdsourcing to solicit ideas from outside (Afuah & Tucci, 2012; Chesbrough, 2003; Piezunka & Dahlander, 2015). Menon et al. (2006) even find that managers are more motivated to learn from external as compared to internal competitors because they pose a lower threat to their personal status.

Thus, evidence on decision-makers' receptivity to externally developed ideas is mixed and spans diverse, often disconnected research streams. In integrating these streams of research, we seek to reconcile findings to shed new light on how managers react to external ideas (Anderson et al., 2014). Drawing on the Dynamic Capabilities Theory (Teece, 2007), we propose a dual-pathway model that highlights the competing risk considerations shaping managers' endorsement of creative ideas from external sources.

1.1. Competing Considerations in Endorsing External Creative Ideas

We define external ideas as knowledge (ideas, processes, technologies) that was originally developed or derived from a source outside the *organizational boundary* (e.g., Antons & Piller, 2015; Katz & Allen, 1982; Rosenkopf & Nerkar, 2001).¹ For example, an employee may propose a creative idea for a product or service inspired by what they have seen in a previous organization or adopt a creative idea they came across when interacting with others, say in an external conference. In both scenarios, the originating source of the idea is external to the focal organization, although it is proposed by an existing employee. Thus, the important feature differentiating internal from external ideas is where the idea originated from or where the core of the idea was originally developed. Although these ideas may require varying degrees of internal adaptation to fit organizational contexts, we classify ideas with external origins as external ideas.

In examining how managers perceive external (compared to internal) creative ideas, we are broadly guided by DCT (Teece, 2007), which provides a framework for understanding how organizations sustain their

competitive advantage in a dynamic business environment. DCT presents two key considerations for firms related to innovation. Firstly, it emphasizes the importance of sensing competitive threats and seizing opportunities to maintain a competitive edge (Teece et al., 1997). Organizations need to stay vigilant to the risk of being outcompeted by market rivals and act promptly to tackle these challenges (Menon & Pfeffer, 2003) by identifying and leveraging emerging opportunities while mitigating potential threats. Secondly, DCT highlights the need to assess if new ideas can be seamlessly integrated with existing systems and processes (Helfat & Peteraf, 2015; Teece, 2018) or if they might be incompatible. Introducing new ideas from outside involves adapting existing systems and procedures to match them, which can be risky and can lead to failure, thus constraining managers from adopting such ideas.

We propose that these organizational-level dynamics manifest at the individual manager-level in terms of managing two kinds of competing risks (Baron, 1996; Ritov & Baron, 1990): the risk of omission (or *not* taking a specific action) versus the risk of commission (or taking action). In our application of DCT to the context of managerial adoption of external innovation, the risk of omission involves managers choosing *not* to adopt or invest in new external ideas, technologies, or strategies, leading the organization to fall behind competitors who are more agile in capitalizing on emerging opportunities (Abrahamson, 1991). Conversely, the risk of commission—the potential negative consequences of taking action—arises when managers opt to integrate external ideas and, as a result, face failure or disruption to existing processes. Thus, we argue that managers experience a tension between the risk of being outcompeted due to not adopting such external ideas and the risk of incompatibility due to implementing them.

1.2. Risk of being Outcompeted

DCT underscores the importance of identifying and capitalizing on new market opportunities or risking the loss of competitive advantage (Teece, 2007; Helfat & Peteraf, 2015). Organizations failing to adopt innovative ideas promptly risk falling behind competitors who implement similar innovations more quickly, affordably, or with superior features (Barney, 1991; Porter, 1980; Wernerfelt, 1984). This highlights the risk of omission, the potential loss of competitive advantage, or market position due to the failure to adopt innovative ideas in a timely manner. In fact, prior research shows that organizations face significant pressure to exploit opportunities quickly, driven by "bandwagon pressures" (Abrahamson, 1991) and concerns that competitors will preempt unexploited opportunities (Hooshangi & Loewenstein, 2016).

Applying this framework, we propose that managers evaluating external creative ideas grapple with the risk of being outcompeted if they delay adoption. We define this risk as the potential loss of competitive advantage due to the failure to implement innovative ideas in time (Teece, 2007). External ideas heighten this perceived risk more than internal ones because they are already partially visible in the public domain. Platforms such as industry conferences, trade shows, and external events expose innovations to a wider audience, making them accessible to competitors (Chesbrough, 2003; Menon & Pfeffer, 2003; West & Bogers, 2014). Unlike internal ideas, which remain largely concealed within the organization, external ideas are more widely known, creating urgency for adoption to maintain competitive positioning (Laursen & Salter, 2006). Taking these arguments together, we posit:

Hypothesis 1a. External (as compared to internal) creative ideas are perceived by managers to involve a higher risk of being outcompeted.

Further, we argue that a high perceived risk of being outcompeted increases managers' willingness to implement creative ideas by allocating resources to bring them to fruition (Zhou et al., 2024). To maintain competitive positioning, decision-makers are more inclined to adopt innovations that provide a unique advantage or counter

¹ The NIH literature suggests that knowledge and ideas can be considered as external when they originate from another organization, a different geographic area or discipline, or domain of expertise. In this paper, we focus on *organizational boundaries* because they have been extensively studied within the knowledge transfer and innovation literature (Argote & Ingram, 2000; Carlile, 2002). Further, prior work has suggested that the impact of knowledge exploration is highest when it spans organizational boundaries (Rosenkopf & Nerkar, 2001).

competitive threats (Lieberman & Montgomery, 1988). In dynamic markets, sensing and seizing opportunities is critical for success (Teece, 2007). Therefore, managers who perceive a high risk of being out-competed by rivals are more likely to recognize the urgency of adopting promising external ideas to avoid falling behind. The fear of loss of competitive advantage can significantly motivate managers to be receptive to external ideas and engage in preemptive adoption of innovations, as neglecting these opportunities can weaken their competitive position, leaving them vulnerable to faster-moving rivals (Ferrier et al., 1999; Hooshangi & Loewenstein, 2016; Schilling, 2002). Thus, we posit:

Hypothesis 1b. Managers' perceived risk of outcompetition associated with a creative idea is positively related to their willingness to implement the idea.

1.3. Risk of Incompatibility with Existing Systems

DCT also highlights that reconfiguring the organization's existing processes and systems to accommodate new ideas is critical to its competitive advantage in dynamic environments (Helfat & Peteraf, 2015; Teece et al., 1997). However, this reconfiguration process is not always straightforward, as it is path-dependent and influenced by the organization's past decisions and investments in operating systems (Arthur, 1989), indicating that even highly innovative ideas may not be worth adopting if they significantly disrupt existing systems and routines. This aligns with the risk of commission or the potential negative consequences of taking action.

Drawing from these principles, we posit that managers will perceive external creative ideas to be associated with a higher risk of incompatibility with existing systems and practices. We define the *perceived risk of incompatibility* as the potential challenges and disruptions anticipated with integrating new ideas into the current organizational routines, systems, and practices (e.g., Teece, 2007; Zahra & George, 2002). External ideas are developed within different organizational contexts and, therefore, may be misaligned or incompatible with the focal organization's established knowledge and operational boundaries (Chesbrough, 2003). Without familiarity with the developmental context of such external ideas (e.g., Cohen et al., 1972), managers may experience uncertainty about whether they would fit with the existing practices of their organization and about the feasibility of idea implementation (Chen et al., 2018). Consequently, they are likely to perceive the adoption of external ideas as potentially requiring significant disruptions to current systems (Helfat et al., 2007; Leonard-Barton, 1992). In contrast, internal ideas are likely more compatible with existing systems and practices, as they emerge from and are shaped by the organization's unique context, resources, capabilities, and culture (Amabile, 1988). Further, internal ideas often build on the organization's existing knowledge base, thus posing a lower risk of incompatibility. Hence, we hypothesize:

Hypothesis 2a. External (as compared to internal) creative ideas are perceived by managers to involve a higher risk of incompatibility with existing systems.

In turn, the perceived risk of incompatibility with existing systems will negatively influence managers' willingness to implement external creative ideas. Given their accountability for both unit and organizational performance, managers may be reluctant to adopt ideas that could (even if only temporarily) impair performance due to compatibility issues (Zahra & George, 2002). Indeed, DCT indicates that an organization's established systems and routines might lead to risk aversion, hindering the exploration and adoption of new opportunities that deviate significantly from current practices (Teece et al., 1997; Leonard-Barton, 1992). This risk of commission associated with external ideas can be paralyzing for managers, as they fear the potential negative consequences of adopting ideas that may then disrupt existing processes

or lead to resource misallocation (e.g., Ritov & Baron, 1990; Kahneman & Tversky, 1982). Thus,

Hypothesis 2b. Managers' perceived risk of incompatibility associated with a creative idea is negatively related to their willingness to implement the idea.

1.4. The Moderating Role of Perceived Technological Turbulence

DCT posits that a firm's ability to reconfigure resources, processes, and routines, seize opportunities, and manage threats (Chen et al., 2018; Teece et al., 1997) is especially vital in rapidly changing technological environments. Technological turbulence refers to the rate and speed of technological changes and innovations in the industry that can lead to unpredictable, and disruptive changes (Eisenhardt & Martin, 2000; Jaworski & Kohli, 1993). Although this has been predominantly studied at the macro level as a given environmental attribute faced by organizations (e.g., Danneels & Sethi, 2011), research suggests that only when decision-makers actively perceive changes in the external environment do these changes influence their decision-making process (e.g., Burke et al., 2006). Thus, technological turbulence is often conceptualized as a socially constructed variable (e.g., Chen et al., 2018), where managers in the same technological environment may perceive different rates of technological change in their industry.

We define *perceived technological turbulence* as managers' subjective assessment of the rate and magnitude of changes in the technological landscape within their industry (Chen et al., 2018). When managers perceive high technological turbulence, they view the immediate environment as characterized by frequent new products, short product development cycles, and rapid technological obsolescence (Atuahene-Gima & Li, 2004; Zhao et al., 2014). These environments offer firms opportunities to gain a competitive edge by adopting innovative ideas and outpacing competitors (Tushman & Anderson, 1986). However, they also increase the difficulty of aligning innovations with existing systems and processes. Consequently, managers' perceptions of technological turbulence can complicate decisions about external ideas by exacerbating the tension between the two types of risk outlined above (Ciborra, 1996).

First, we argue that managers' perceptions of technological turbulence amplify the perceived risk of being outcompeted when considering external ideas. When managers perceive their environment as highly turbulent, they feel increased pressure to stay attuned to external developments. In their view, external ideas, being more visible and accessible to other companies, appear particularly vulnerable to quick adoption by rivals, thereby giving them a competitive edge (Chesbrough, 2003; Lieberman & Montgomery, 1988). This perceived visibility and the shorter time window for capitalizing on ideas in turbulent environments thus make external ideas feel riskier to not endorse (Eisenhardt & Martin, 2000). In contrast, internal ideas remain relatively protected from this external scrutiny, reducing the risk of immediate imitation by competitors (Giarratana & Mariani, 2013), thus reducing managers' anxiety about losing to rivals. When they perceive technological turbulence to be lower, the competitive landscape feels more predictable. Thus, managers' sense of urgency around external ideas diminishes as competitors are less likely to rapidly adopt new technologies, thereby reducing the relative advantage of acting on external ideas. Thus, we argue:

Hypothesis 3a. The impact of external (vs. internal) creative ideas on managers' perceived risk of being outcompeted will be moderated by managers' perceived technological turbulence such that this relationship will be stronger when turbulence is higher (vs. lower).

Next, we hypothesize that managers' heightened perception of technological turbulence amplifies the perceived incompatibility of external ideas relative to internal ideas. When managers perceive high technological turbulence, they anticipate that it will be difficult for their

organizations to adapt processes to accommodate external ideas. Organizations develop unique processes and systems that constrain their flexibility, thereby making the adoption of new, external ideas a challenge (Teece et al., 1997; Leonard-Barton, 1992). When managers perceive more turbulence, they may believe that the speed of technological change outpaces their organization's ability to reconfigure its systems, highlighting the challenges of integrating new ideas that may quickly become outdated or obsolete (Henderson & Clark, 1990). Managers may worry that novel external ideas will introduce additional layers of complexity, making it difficult to ensure compatibility and interoperability with existing systems. In contrast, when perceived turbulence is low, their perception of a slower pace would suggest a better potential for compatibility, diminishing the perceived gap between external and internal ideas. Thus,

Hypothesis 3b. The impact of external (vs. internal) creative ideas on managers' perceived risk of incompatibility will be moderated by perceived technological turbulence such that this relationship would be stronger when turbulence is higher (vs. lower).

Together, H1a-b and 3a and H2a-b, and 3b, respectively, propose mediated moderation relationships wherein the indirect effects of idea source on idea implementation decisions via the two risk mechanisms are moderated by perceived technological turbulence. Thus, we propose:

Hypothesis 4a. The positive indirect effect of external (vs. internal) creative ideas on managers' willingness to implement such ideas via the perceived risk of being outcompeted will be moderated by managers' perceived technological turbulence such that this indirect effect will be stronger when turbulence is higher (vs. lower).

Hypothesis 4b. The negative indirect effect of external (vs. internal) creative ideas on managers' willingness to implement such ideas via perceived risk of incompatibility will be moderated by managers' perceived technological turbulence such that this indirect effect will be stronger when turbulence is higher (vs. lower).

2. Overview of studies

In testing our hypotheses, we conducted four studies—a pilot study, a quasi-experimental field study, and two experimental studies.² In the pilot study, our aim was to identify the considerations that are salient for decision-makers in real organizations when evaluating creative ideas and to establish the external validity of the theorized risk perceptions regarding external vs. internal ideas (H1a, H2a). Study 1, a pre-registered experimental study, was designed to demonstrate the incremental validity of our proposed mediators (H1a-b, H2a-b) while controlling for alternative explanations suggested by prior work. Next, we conducted Study 2, a pre-registered quasi-field experiment, with managers evaluating specific ideas. Finally, Study 3, a pre-registered experimental study with working professionals, tested our full model. All studies were approved by the Institutional Review Board at our university (Protocol #1608959). All materials are available on an OSF webpage. Data, codes for analyses, and pre-registration documents for all studies are available on the dedicated OSF webpage (https://osf.io/quhv3/?view_only=61c12125271e4d77b68eeda81d79c4bc). Data for the field study are not posted due to intellectual property concerns underlying idea-level data, but the syntax used is posted.

2.1. Pilot study: method

We first conducted a pilot study to establish the external validity of

our proposed risk-related reasons as well as explore other considerations managers grapple with when evaluating ideas and how these considerations differ for external ideas versus internal ideas. We recruited 154 U. S. participants from Prolific with at least 3 years of managerial experience in evaluating employees' ideas for new products and services, offering a compensation of \$4.00 for a twenty-minute session. Out of 154 participants, 33 participants did not complete the survey. Of the final sample of 121 participants, 61 % were male, and 49.4 % were Caucasian. The average age was 44.7 years ($SD = 11.24$), and the average management experience was 10.5 years ($SD = 6.76$).

Adopting a mixed qualitative-quantitative study design, we asked participants to recall a time when one of their employees approached them with a significant creative idea that they wanted to implement. Participants then detailed the source and origin of the idea and described the considerations they used in evaluating it. For the quantitative component, participants answered some survey questions about their recalled ideas and answered demographic questions.

Idea Source. We coded each idea's source (internal = 0 and external = 1) based on participants' responses to whether the idea was originally developed inside or outside their organization. Out of 121 participants, 38 provided external ideas, and 83 provided internal ideas.

2.2. Qualitative responses coding

We asked participants to write down the criteria or considerations they used to evaluate the idea. We then provided labels and definitions of different potential categories to two coders, who categorized the responses. The categories were determined by our own theory as well as prior research examining the adoption of external ideas (e.g., Antons & Piller, 2015). Initially, the two coders applied the preliminary coding scheme to 20 % of the sample and then met to resolve any discrepancies (e.g., Parke et al., 2022). Responses broadly fell into five of the provided categories: (i) "risk of incompatibility," related to the fit of the proposed idea with existing practices and systems; (ii) "competitive advantage," or how the proposed idea could impact the company's competitive standing—this included responses relating to the potential loss of competitive advantage if the idea were not adopted and potential gains if it were adopted; (iii) "misalignment with organizational values," or how well the proposed idea aligns with the organization's values; (iv) "organizational image concerns," about protecting the organization's image or reputation; and (v) "cost," pertaining to expense-related concerns. Responses did not address other categories, such as the "risk of losing status" (Menon & Pfeffer, 2003) or the risk of "being perceived as idea theft" (Ellis, 2022), that have been highlighted in other research.

2.3. Quantitative measures

To further investigate the initial validity of our two proposed mediators—the risk of incompatibility and being outcompeted—beyond the reasons suggested by prior research, we also assessed various affective and cognitive mechanisms indicated by our pre-defined categories using survey items. Participants rated the measures on a 5-point Likert-type scale (1 = *Strongly Disagree* to 5 = *Strongly Agree*). All scale items are included in Online Appendix 1.

Risk of Outcompetition. Participants rated three items adapted from the competitive advantage of technology application scale (Sethi & King, 1994). The items are: "Our competitors can adapt this idea first, risking a loss of competitive advantage for us," "Our competitors can beat us to it, and we risk falling behind," and "Our competitors may overtake our position in the market if we don't adopt this idea" ($\alpha = 0.89$).

Risk of Incompatibility. We developed three items based on our definition of risk incompatibility. The items are: "Adopting such an idea would run the risk of not fitting with our existing practices," "Implementing such an idea would carry the risk of disrupting or requiring significant changes in our existing practices," and "Adopting this idea

² We also conducted additional studies. Details of these studies, additional analyses, extended write-ups (where applicable), and study materials are provided in the Online Appendix on the OSF webpage (https://osf.io/quhv3/?view_only=61c12125271e4d77b68eeda81d79c4bc).

carries the risk of not being relevant to my organization” ($\alpha = 0.76$).

Organizational Image Concerns. Using the corporate image and reputation scale (Nguyen & Leblanc, 2001) as a guide, we created three items to capture organizational image concerns (e.g., “Using this idea would reflect badly on my organization’s image” ($\alpha = 0.96$).

Misalignment with Organizational Values. As managers frequently serve as gatekeepers who preserve the culture and values of the team and organization, they may perceive external ideas as inconsistent with these values. Using the in-group protectionism scale (Smeekees & Verkuyten, 2013) as a guide, we created three items to measure this construct. A sample item is: “Using this idea would risk not fitting with my organizational values” ($\alpha = 0.95$).

Risk of Losing Status. Managers’ fear of losing esteem, respect, or perceived competence by advocating for external ideas (Menon et al., 2006) was measured with three items, adapted from the status threat measure (Harari, Parke, & Marr, 2022). A sample item is: “If I implement this idea in my organization, my competence would be questioned” ($\alpha = 0.95$).

Cost. Managers’ perceptions of the potential costs associated with implementing external ideas were measured using three items. One of these items is: “Implementing this idea will be too costly for my organization” ($\alpha = 0.94$).

Risk of Idea Theft. We used three items to measure intellectual property concerns (e.g., Ellis, 2022). An item is: “Accepting this idea runs the risk of stealing others’ ideas” ($\alpha = 0.91$).

3. Pilot study: results and discussion

3.1. Qualitative findings

Table 1 summarizes representative quotes from the qualitative data. A majority of the sample mentioned considering factors related to the risk of incompatibility (81.0 %), competitive advantage (78.5 %), cost (60.3 %), and misalignment with organizational values (47.9 %). 2.5 % expressed organizational image concerns; other categories were not mentioned.

3.2. Preliminary hypotheses testing

We then used responses to the measures on risk perceptions about external versus internal ideas to explore H1a and H2a. H1a, which proposed that ideas from external as compared to internal sources will be associated with a higher risk of being outcompeted, was supported ($M_{\text{external}} = 3.09, SD = 1.01; M_{\text{internal}} = 2.53, SD = 1.09; \Delta M = 0.56, t(119) = 2.66; \text{Cohen's } d = 0.52, 95 \% \text{ CI } [.13, 0.91]$). H2a, which proposed that ideas from an external source would entail a greater risk of incompatibility, was also supported ($M_{\text{external}} = 2.86, SD = 1.12; M_{\text{internal}} = 2.12, SD = 1.03; \Delta M = 0.74, t(119) = 3.57; p < 0.001; \text{Cohen's } d = 0.70, 95 \% \text{ CI } [.31, 1.09]$).

3.3. Supplementary Analyses: Alternative mediators

Although we theorized managers’ perceived risk of incompatibility and outcompetition as the two main concerns about external ideas, we also explored whether alternative reasons suggested by prior work were supported. To establish the discriminant validity of our measures, we first conducted a confirmatory factor analysis. The seven-factor full model, in which the seven variables measured above were treated as separate factors, demonstrated a reasonably good fit to the data ($\chi^2(175) = 361.49, p < 0.01; \text{RMSEA} = 0.09; \text{CFI} = 0.93; \text{TLI} = 0.92; \text{SRMR} = 0.08$). This measurement model provided a better fit compared to other models where different measures were combined. Details of model fit are available in Online Appendix 2.

First, in examining whether the idea source also affected the above-mentioned alternative mediators, we conducted independent sample t-tests. Results indicated that the idea source had a significant effect on the

Table 1
Managers’ risk considerations for idea implementation.

Risk considerations	Representative Quotes
Risk of incompatibility	“I had to definitely consider disruption to our company’s existing banking collections and disbursements. Had to make sure not to disrupt company operations.” 81.0 % of the participants stated that they consider risks that the idea will not fit the existing practices or system. “I saw it as a supplemental thing not something that would interrupt our normal business.” “It could cause disruption or confusion to the teams and employees already existing ways of doing things.”
Competitive advantage	“If we had failed to act quickly, our competitor would have gained the upper hand.”“Other companies were already implementing this process, so adopting it made sense to not lost competition” “Relating to the market, this would give us an advantage over majority of the competition in the industry.” “The idea would give advantages against competitors and some other organizations were already doing it.” “The overall time cost to get an alpha implemented was the largest issue.”
Cost	60.3 % of the participants expressed that they consider risks associated with the idea being too costly. “Is the cost beneficial to the bottom line of production? Is there a cost savings or lose due to this production process?” “There was pushback from many groups. My employee did an excellent job in laying out the potential cost savings and increased quality of implementing a unified accounting system.”
Misalignment with organizational values	“I assessed how well the idea aligned with our organizational goals, values, and strategic direction.” “I am confident when implemented, the idea will fit into the organization values and goals.” “The organization is in the business of doing things along the lines of this idea, so it definitely meshed with the values and trends of the company.”
Organizational image concerns	“We wanted better reputation as an innovative company.” “As a VP Finance and Treasurer, it’s my job to protect the company’s image.”

risk of losing status ($M_{\text{external}} = 2.16; M_{\text{internal}} = 1.66; \Delta M = 0.50; t(119) = 2.55; p = 0.012; d = 0.50, 95 \% \text{ CI } [.11, 0.89]$), cost ($M_{\text{external}} = 2.82; M_{\text{internal}} = 2.30; \Delta M = 0.52; t(119) = 2.15; p = 0.034; d = 0.42, 95 \% \text{ CI } [.03, 0.81]$), and idea theft ($M_{\text{external}} = 2.04; M_{\text{internal}} = 1.63; \Delta M = 0.41; t(119) = 2.15; p = 0.034; d = 0.42, 95 \% \text{ CI } [.03, 0.8]$). However, the idea source did not have a significant effect on organizational image concerns ($M_{\text{external}} = 1.90; M_{\text{internal}} = 1.65; \Delta M = 0.26; t(119) = 1.22; p = 0.226; d = 0.24, 95 \% \text{ CI } [-0.15, 0.62]$) or misalignment with organizational values ($M_{\text{external}} = 2.45; M_{\text{internal}} = 2.02; \Delta M = 0.42; t(119) = 1.83; p = 0.07; d = 0.36, 95 \% \text{ CI } [-0.03, 0.74]$).

3.4. Pilot study discussion

Using a recall task, this pilot study aimed to identify mechanisms that are both salient to decision-makers (based on spontaneous mentions in qualitative comments) and that differentiate between internal and external ideas. Our analysis provided support for the external validity of our two proposed factors for why external ideas diverged significantly from internal ones: the risk of incompatibility with existing systems and the risk of being outcompeted in the market. External ideas were also perceived as more costly. While not salient in the qualitative comments, external ideas were also perceived to involve a greater risk of losing status and idea theft in the survey measures. However, the uneven sample distribution of external versus internal ideas (fewer external ideas were shared) may have influenced these comparisons. Therefore, we conducted Study 1, an experimental study, to more comprehensively examine these mechanisms and establish the incremental validity of our proposed mechanisms.

3.5. Study 1: Method

A power analysis conducted with G*Power, assuming an alpha level of 0.05, power of 0.8, and a small-to-medium effect size of 0.2 (Cohen, 1988), indicated that 199 participants were required for our final sample. Considering the typical high attrition rates in online samples and the possibility of incomplete responses or failure to pass attention checks (Meade & Craig, 2012), we recruited 251 U.S. participants from Prolific, offering \$1.20 each for a 5-minute survey. This study was pre-registered (AsPredicted #163510). We specifically recruited managers with a minimum of 2 years of managerial experience. We excluded five participants who failed an attention check and two who did not complete the survey. In the final sample of 244 respondents, 52.2 % were male, and 65.2 % were Caucasian. The average age was 43.1 years ($SD = 11.94$), and the average management experience was 10.2 years ($SD = 7.90$).

Using a 2-condition (idea source: external vs. internal) between-subjects design, we randomly assigned participants to one of the two conditions. Participants were asked to imagine they were managers in the customer service division of an electronics company. We provided information about the company and its current, conventional approach to customer service, which included an automatic call distribution system. We then introduced a new idea proposed by an employee, Taylor, to integrate AI-powered remote visual assistance technology into the customer service system.³ Following this, we administered the idea source manipulations and then measured the mediators, the dependent variable, manipulation checks, and demographics.

3.6. Manipulations

Idea source manipulation. In line with our definition of an external source as an idea originating from outside (versus inside) the organization's boundary (Antons et al., 2017; Antons & Piller, 2015), participants in the external source condition ($N = 125$) were informed that the technology proposed by the employee, Taylor, was originally developed by an external research think tank. In contrast, those in the internal source condition ($N = 119$) were told that the technology was initially developed by an internal research task force within their own organization. Participants in the external idea source condition read:

Taylor proposed that we consider the use of a new technology developed by an external research think tank. Taylor had read a recent report released by this external research institute detailing a new technology they had developed, which could be potentially applied to different situations. Taylor felt that this technology, which originates

from an external source outside your organization, could be a viable solution capable of addressing your current challenges and capturing potential benefits.

Participants in the internal idea source condition read:

Taylor proposed that we consider the use of a new technology developed by your company. Taylor had read a recent report released by your company's research taskforce detailing a new technology that had been developed internally, which could be potentially applied to different situations.

Taylor felt that this technology, which originates internally within your organization, could be a viable solution capable of addressing your current challenges and capturing potential benefits.

3.7. Measures

We conducted a scale validation study based on the guidelines provided by Colquitt et al. (2019) to establish the validity of our focal mediators (see Online Appendix 4). Participants used a 7-point Likert-type scale (1 = *Strongly Disagree* to 7 = *Strongly Agree*) to rate all measures.

Risk of Outcompetition. This was rated using the three items in the pilot study ($\alpha = 0.75$).

Risk of Incompatibility. Participants rated the three items from our pilot Study ($\alpha = 0.82$).

Willingness to Implement Idea. This was assessed using Baer's (2012) three-item measure. A sample item is, "I will provide resources (e.g., funds, materials, staffing) for the development of this idea" ($\alpha = 0.92$).

Organizational Image Concerns. We used the three items from the pilot Study ($\alpha = 0.96$).

Misalignment with Org. Values. We used the three items in the pilot Study ($\alpha = 0.92$).

Risk of Idea Theft. We used the same three items from the pilot Study ($\alpha = 0.94$).

Cost. We used the same three items as in the pilot Study to measure this ($\alpha = 0.94$).

Risk of Losing Status. We measured this with three items as in the pilot Study ($\alpha = 0.94$).

Perceived Idea Maturity. To assess the possibility that external ideas might be perceived to be further along in development, we used three items developed for this study. A sample item is: "The idea has been fully developed" ($\alpha = 0.93$).

Perceived Expertise of Idea Source. To measure if managers perceived internal idea sources as having more expertise and credibility, we adapted three items from the perceived expertise scale (Rentz et al., 2002; e.g., "The source of Taylor's idea is credible" $\alpha = 0.90$).

Perceived Familiarity with Idea. To assess the employee Taylor's perceived familiarity and understanding of the idea, we adapted three items from the perceived familiarity measure (Greul et al., 2023). A sample item is: "This employee is familiar with the idea" ($\alpha = 0.93$).

Perceived Idea Support. To examine differences in the level of support expected from others for the idea, we adapted three items from Venkataramani et al. (2023). A sample item is: "I think other members in the organization will help with implementing this idea" ($\alpha = 0.93$).

4. Study 1: Results and discussion

4.1. Manipulation checks

Participants responded to two items each regarding the origin of the employee's idea. Sample items are: "The technology suggested by the employee was originally developed within my company" (for internal source) and "The technology suggested by the employee was originally developed outside my company" (for external source). Participants in the internal condition ($M = 6.05$, $SD = 1.41$) perceived the idea as internally sourced compared to those in the external condition ($M =$

³ A separate pilot study established that this idea was considered highly novel and useful, in other words, creative.

Table 2
Correlations and Descriptive Statistics (Study 1).

	<i>M</i>	<i>SD</i>	1	2	3	4
1. Idea Source ^a	0.49	0.50				
2. Risk of being Outcompeted	4.78	1.24	0.26**	(0.75)		
3. Risk of Incompatibility	3.59	1.38	0.39**	0.29**	(0.82)	
4. Willingness to Implement	5.74	0.92	-0.14*	0.17**	-0.40**	(0.92)

Notes: *N* = 244. Alpha reliabilities are presented along the diagonal. ^a Idea source (0 = internal, 1 = external). ^b $p < 0.10$; * $p < 0.05$; ** $p < 0.01$.

1.94, $SD = 1.50$; $F(1,242) = 444.31$, $p < 0.001$, $\eta_p^2 = 0.65$). Similarly, participants in the external condition ($M = 6.37$, $SD = 1.02$) identified the idea source as external compared to those in the internal condition ($M = 2.35$, $SD = 1.83$; $F(1,242) = 485.99$, $p < 0.001$, $\eta_p^2 = 0.67$). These results confirmed the effectiveness of our manipulations.⁴

4.2. Hypothesis testing

Table 2 provides the descriptive statistics and correlations among core study variables, and Table 3 provides the results of the regression analyses. Online Appendix 2 provides the results of our confirmatory factor analysis to establish the discriminant validity of all our scale measures.

One-way ANOVA results indicated that managers' implementation willingness for external ideas was significantly lower than that for internal ideas ($M_{\text{external}} = 5.60$, $SD = 0.97$; $M_{\text{internal}} = 5.86$, $SD = 0.86$; $\Delta M = 0.26$, $F(1,242) = 4.91$, $p = 0.028$, $\eta_p^2 = 0.02$). Supporting H1a, external ideas were significantly higher in perceived risk of being outcompeted than internal ones ($M_{\text{external}} = 5.11$, $SD = 1.12$; $M_{\text{internal}} = 4.46$, $SD = 1.28$; $\Delta M = 0.65$, $F(1,242) = 17.99$, $p < 0.001$, $\eta_p^2 = 0.07$). Further, supporting H2a, external ideas were perceived as more incompatible than internal ones ($M_{\text{external}} = 4.14$, $SD = 1.27$; $M_{\text{internal}} = 3.07$, $SD = 1.27$; $\Delta M = 1.07$, $F(1,242) = 43.15$, $p < 0.001$, $\eta_p^2 = 0.15$).

Next, supporting H1b, the regression of willingness to implement on the perceived risk of outcompetition showed a significant positive relationship ($B = 0.23$, $p < 0.001$). We then tested the mediation model implied by H1a-b using the PROCESS macro (model 4; 5000 bootstrapping iterations; Hayes, 2013). The results showed a significant indirect effect of external idea source on implementation willingness through managers' perceived risk of being outcompeted (estimate = 0.15, $SE = 0.05$, 95 % CI [0.07, 0.25]). Similarly, supporting H2b, perceived risk of incompatibility was significantly negatively related to willingness to implement ($B = -0.33$, $p < 0.001$). Indirect effect analyses further indicated a significant indirect effect of external idea source on implementation willingness via perceived risk of incompatibility (estimate = -0.34, $SE = 0.07$, 95 % CI [-0.49, -0.22]). Table 4 provides the direct, indirect, and total effects.

Alternative Mediators. Next, we examined whether our idea source manipulation affected the alternative mediators. Independent sample *t*-tests indicated that idea source had a significant effect on the risk of misalignment with organizational values ($M_{\text{external}} = 3.23$; $M_{\text{internal}} =$

2.57; $\Delta M = 0.66$; $F(1,242) = 15.32$, $p < 0.001$, $\eta_p^2 = 0.06$) and cost ($M_{\text{external}} = 3.95$; $M_{\text{internal}} = 3.39$; $\Delta M = 0.56$; $F(1,242) = 9.61$, $p = 0.002$, $\eta_p^2 = 0.04$), but not on organizational image concerns ($M_{\text{external}} = 2.23$; $M_{\text{internal}} = 2.05$; $\Delta M = 0.18$; $F(1,242) = 1.45$, $p = 0.23$, $\eta_p^2 = 0.01$), perceived idea theft ($M_{\text{external}} = 2.64$; $M_{\text{internal}} = 2.35$; $\Delta M = 0.29$; $F(1,242) = 2.67$, $p = 0.10$, $\eta_p^2 = 0.01$), or risk of losing status ($M_{\text{external}} = 1.71$; $M_{\text{internal}} = 1.59$; $\Delta M = 0.11$; $F(1,242) = 0.90$, $p = 0.35$, $\eta_p^2 = 0.004$).

In testing whether the two focal mediators (risk of incompatibility and risk of outcompetition) predicted implementation willingness above and beyond these alternative mediators, we utilized the PROCESS macro, including these two mediators while controlling for other alternative mechanisms. The results showed that the indirect effect of the external idea source on implementation willingness via perceived risk of incompatibility (estimate = -0.07, $SE = 0.03$, 95 % CI [-0.15, -0.02]) and via perceived risk of being outcompeted (estimate = 0.08, $SE = 0.03$, 95 % CI [0.03, 0.15]) remained significant. Regression results are provided in Online Appendix 2.

4.3. Study 1 discussion

In Study 1, we examined the effects of the manipulated idea source on our two primary mediators, along with several alternative mediators, and managers' willingness to implement these ideas. We demonstrate that external ideas carry a higher perceived risk of incompatibility and risk of being outcompeted compared to internal ideas for managers. Moreover, the perceived risk of incompatibility negatively impacts implementation willingness, while the perceived risk of outcompetition positively influences it. We further show that these two primary mediators exceed other explanations in clarifying the effects of idea source on implementation willingness. Although Study 1 has strengths related to internal validity, we sought to establish the external validity of our findings. Therefore, we conducted Study 2 in a field setting.

5. Study 2: Method

Study 2 was a pre-registered (AsPredicted #163158) quasi-experimental field study that tested our model using data collected from a major commercial bank in China. Our interviews with top management indicated that customers in the banking industry generally expected novel products and services and could switch banks based on these offerings. Technological advancements have empowered customers to conduct banking activities anytime, anywhere, using various digital channels such as mobile apps and online banking platforms. Further, the rise of fintech startups and the use of artificial intelligence, blockchain, and data analytics presents both opportunities and challenges for banks. These technologies have the potential to streamline operations, improve risk management, and enhance customer experiences. Moreover, depending on their specific domain, managers experienced differences in the degree of dynamism of their technical environment. Thus, this provided a good setting to test our ideas.

5.1. Study 2 pilot task

To determine how managers rated ideas objectively based on their

⁴ In line with our pre-registration, we also examined if our manipulations affected the other variables. However, there was no difference in perceived idea maturity between idea sources ($M_{\text{external}} = 4.20$; $M_{\text{internal}} = 4.22$; $\Delta M = 0.02$; $F(1,242) = 0.02$, $p = 0.90$, $\eta_p^2 = 0.00$). Significant differences were found between idea source regarding the expertise of the idea source ($M_{\text{external}} = 5.38$; $M_{\text{internal}} = 5.70$; $\Delta M = 0.32$; $F(1,242) = 8.13$, $p = 0.005$, $\eta_p^2 = 0.03$), familiarity with the idea ($M_{\text{external}} = 5.44$; $M_{\text{internal}} = 5.73$; $\Delta M = 0.29$; $F(1,242) = 6.16$, $p = 0.014$, $\eta_p^2 = 0.03$), and idea support ($M_{\text{external}} = 5.47$; $M_{\text{internal}} = 5.73$; $\Delta M = 0.26$; $F(1,242) = 5.21$, $p = 0.023$, $\eta_p^2 = 0.02$). Nevertheless, even after controlling for these variables (e.g., attributes associated with the idea source), the risk of outcompetition ($B = 0.13$, $p < 0.001$) and the risk of incompatibility ($B = -0.16$, $p < 0.001$) remained significant predictors of implementation willingness.

Table 3
Unstandardized Results of Regression Analysis (Study 1).

	Model 1 Risk of Outcompetition		Model 2 Risk of Incompatibility		Model 3 Willingness to Implement		Model 4 Willingness to Implement	
	B	SE	B	SE	B	SE	B	SE
Idea Source ^a	0.65***	0.15	1.07***	0.16	−0.26*	0.12	−0.07	0.11
Risk of Outcompetition							0.23***	0.04
Risk of Incompatibility							−0.32***	0.04
R ²	0.07		0.15		0.02		0.25	

Notes: N = 244; table entries represent unstandardized parameter estimates. ^aIdea Source (External = 1, Internal = 0). [†]p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

Table 4
Direct, Indirect, and Total Effects (Study 1).

DV	IV	Direct Effect	Indirect effect	Total Effect
Willingness to Implement Idea	Idea Source	−0.07	−0.19*	−0.26*
	Risk of Outcompetition	0.23***	−0.34**	
	Risk of Incompatibility	−0.32***	0.15***	
Risk of Outcompetition	Idea Source	0.65***		
Risk of Incompatibility	Idea Source	1.07***		

†p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001.

sources, we first conducted a pilot study to identify new ideas before having managers evaluate them. We then manipulated the source of each idea in our main study, the quasi-experiment. For the pilot study, guided by HR's recommendation, we invited 11 employee representatives from sales, marketing, and other customer-facing departments to join an idea-generation group discussion. They were informed that the company was seeking creative ideas for products, marketing services, and process optimization. First, these employees were asked to take a few days to think about potential ideas individually and submit them through an online survey platform, which our HR liaison helped consolidate into 13 unique ideas.

We then hosted a two-hour focus group with the original idea generators to further refine these ideas, discuss their pros and cons, and select a smaller set of unique ideas for our main study. During this process, we removed all identifying information about the original idea generators and assured participants that each developer's identity would remain confidential. We emphasized that the sole goal of the discussion was to assess and improve each idea based on its own merits. We then asked participants to merge similar or compatible ideas, ultimately consolidating them into seven distinct ideas for our main study.

5.2. Main Study

Study 2 involved 183 out of 204 managers (89.71 %) at the same bank who routinely approved or rejected new ideas from their direct reports and thus were appropriate participants for this study. They were provided with the seven ideas from the pilot study and asked to assess each idea on its perceived risk of incompatibility, being outcompeted by competitors, and willingness to implement it. Data collection occurred at two time points to reduce fatigue: demographic and control variables at Time 1, followed by idea evaluations at Time 2 (one week later). When presenting each idea (in randomized order), we manipulated the idea source by randomly attributing them to either an internal or external source. Surveys were translated from English to Chinese using standard translation-back translation procedures (Brislin, 1980). At T1, as part of an event with all managers, one of the coauthors explained the broad research and protocol and assured them of the confidentiality of responses and that the survey was voluntary.

Of the 183 managers who agreed to participate, 17 submitted incomplete surveys at T1 and were consequently removed from further analyses, leading to a response rate of 90.7 % at T1. At T2, 150 out of the 166 T1 manager-participants provided ratings for the seven ideas, achieving a response rate of 82.0 % of the initial participants. After matching all the idea ratings from the two time points and across

different raters, our final sample comprised 1,025 ratings related to 7 new product ideas from 150 managers. Among these managers, 69.3 % were female, 97.3 % held a college degree or higher, and their average tenure was 11.3 years (*SD* = 10.01).

5.3. Idea source manipulation

To manipulate the idea source, we randomly assigned each of the ideas for each manager to either an internal or external source condition. Thus, the same idea was seen as originating internally by some managers but as originating externally by others. Before each idea, in the external source condition, managers were told:

The following idea, which was suggested by an employee at your organization, originated from outside your organization. That is, this idea was originally developed by those working at a different organization, which the employee feels can be implemented in your organization.

In the internal source condition, they were told:

The following idea, which was suggested by an employee, originated from inside your organization. That is, this idea was originally developed by those working within your own organization, which the employee feels can be implemented in your organization.

Measures

Unless otherwise indicated, all items (provided in Online Appendix 1) were rated using a 7-point Likert scale ranging from 1 = 'Strongly Disagree' to 7 = 'Strongly Agree'.

Idea Source. We dummy-coded our manipulation of the idea source so that it assigned a value of 1 for the external source condition and 0 for the internal source condition.

Risk of Outcompetition. At Time 2, managers in our sample rated each of the seven ideas on the extent to which they could be outcompeted by their competitors if they did not implement that idea. We used the same three items from Study 1 (α = 95).

Risk of Incompatibility. Similar to the above, at Time 2, managers rated each idea using the three-item scale used in Study 1 (α = 95).

Willingness to Implement Idea. We measured managers' willingness to implement each idea using Baer's (2012) three-item measure as in Study 1 (α = 94).

Technological Turbulence. At Time 1, we measured managers' perceived technological turbulence at Time 1 using Jaworski and Kohli's (1993) three-item scale. A sample item is, "The technology in our industry is changing rapidly;" (α = 88).

Controls. Although our hypotheses were at the idea level, ratings were nested within managers. Therefore, we controlled for some

manager-related variables that have been shown by prior research to affect their willingness to implement new ideas. For example, based on prior work that has shown that demographic factors can influence reactions to new ideas (e.g., Zhou et al., 2017), we controlled for managers' age, gender, and education (college degree = 1, no college degree = 0). Next, we controlled for managers' openness to experience, a trait that has been shown to be important in affecting idea assessment (Mueller et al. 2012), using the 6-item scale from George and Zhou (2001). A sample item was "I enjoy hearing new ideas" ($\alpha = .77$).

Further, given the variation in our 7 ideas, we controlled for certain idea-level characteristics that can systematically vary with the independent variable (idea source) and dependent variable (implementation willingness). Specifically, we controlled for each idea's novelty and usefulness, as well as whether it aligned with the overall strategic goals of the company. Four members of the top management team served as subject matter experts (SMEs; Hennessey & Amabile, 2010) and rated each idea on these variables. Novelty was measured using the five-item novelty scale from Berg (2016) (e.g., "This idea is novel"; $\alpha = .98$). Usefulness was measured using four items from Amabile (1988) (e.g., "This idea is useful"; $\alpha = .97$). Strategic alignment was measured using three items from Gerow et al. (2015) (e.g., "This idea aligns with our business's strategic plan"; $\alpha = .95$). Following prior work (e.g., Berg, 2016; Zhou et al., 2017), we used R_{wg} and average deviation (AD) indices to calculate agreement across the rater scores (Mean $AD_{Novelty} = 0.27$, $R_{wg} = 0.93$; Mean $AD_{Usefulness} = 0.26$, $R_{wg} = 0.91$; Mean $AD_{Alignment} = 0.31$, $R_{wg} = 0.92$). All R_{wg} values were above 0.70, and AD values were less than the 0.8 upper-limit suggested by LeBreton and Senter (2008), indicating high agreement. Therefore, we averaged these ratings to generate scores of idea novelty, usefulness, and strategic alignment.

Analytical approach

Our data involve two hierarchical levels (ideas nested within managers). Hence, variables related to ideas (such as the manipulated idea source, idea novelty, usefulness, and strategic alignment) were Level 1 cases nested within managers as raters (Level 2). Person-related variables such as managers' demographics, personality, and other perceptions (technological turbulence) were entered as predictors at Level 2. Our mediators (risk of being outcompeted, risk of incompatibility) and DV (willingness to implement) were allowed to vary at both levels. Intraclass correlation coefficients (ICCs) indicated significant between-rater effects for both the mediators and the dependent variable (ICC Outcompetition risk = 0.52; ICC Incompatibility risk = 0.58; ICC implementation = 0.49). Accordingly, we tested our hypotheses using a multilevel moderated mediation model where all paths were estimated simultaneously, using Mplus 8 (Muthén & Muthén, 2018). Following the recommendations of Enders and Tofghi (2008), we group mean-centered our Level 1 predictors and grand mean-centered between-person factors.

6. Study 2: Results and discussion

6.1. Manipulation checks

After each idea, to assess if managers correctly identified the manipulated source, we asked them whether the idea was sourced internally (coded as 0) or externally (coded as 1). We then conducted a mixed-effects logistic regression predicting managers' perceived idea source by the actual idea source, accounting for idea and manager fixed effects. Results revealed a significant effect of idea source on the perceived source, $F(1, 869) = 62.11$, $p < .001$, indicating that participants were significantly more likely to perceive an idea as external when it was presented as such. When ideas were external, participants were 181 times more likely to identify them as such ($B = 5.20$, $SE = 0.66$, $OR = 181.09$, 95 % CI [49.61, 661.01]).

6.2. Hypothesis testing

Descriptive statistics and correlations among all study variables are presented in Table 5. We conducted CFAs to examine the distinctiveness of our focal variables. CFA results indicated that a three-factor model was a better fit to the data ($\chi^2(24) = 151.99$, $p < .01$; RMSEA = 0.072; CFI = 0.98; TLI = 0.97; SRMR = 0.04) than a model where both types of risk were combined ($\chi^2(25) = 1461.13$, $p < .01$; RMSEA = 0.24; CFI = 0.79; TLI = 0.69; SRMR = 0.13) or when all focal variables were combined ($\chi^2(27) = 3937.74$, $p < .01$; RMSEA = 0.38; CFI = 0.42; TLI = 0.22; SRMR = 0.24). The unstandardized results of our path model are presented in Table 6.

First, the manipulated idea source, while in the expected direction, did not have a significant main effect on managers' willingness to implement each idea ($\gamma = -0.07$, $SE = 0.05$, $p = 0.16$). H1a, which predicted that external ideas would have a higher risk of outcompetition, was not supported (Table 6; $\gamma = 0.04$, $SE = 0.06$, $p = 0.56$). H2a, that external ideas would be perceived to increase incompatibility risk, was not supported ($\gamma = 0.07$, $SE = 0.05$, $p = 0.19$).

H3a predicted that the relationship between external idea source and perceived risk of being outcompeted would be moderated by managers' perceived technological turbulence, such that this relationship would be stronger when the perceived turbulence is higher. The cross-level moderating effect of technological turbulence on the relationship between idea source and risk of outcompetition was significant ($\gamma = 0.11$, $SE = 0.05$, $p = 0.02$), supporting H3a. Simple slopes analyses indicated that this relationship was positive and significant ($\gamma = 0.17$, $SE = 0.08$, $p = 0.03$) at high levels (+1 SD) of turbulence but not significant at low levels (-1 SD) of turbulence ($\gamma = -0.10$, $SE = 0.10$, $p = 0.30$; $\Delta\gamma = 0.27$, $SE = 0.12$, $p = 0.02$). This interaction is illustrated in Fig. 2.

Similarly, H3b predicted that the relationship between external idea source and perceived risk of incompatibility would be moderated by the perceived technological turbulence such that this relationship would be stronger when the perceived turbulence is higher. The cross-level moderating effect of technological turbulence on the relationship between idea source and risk of incompatibility was positive and significant ($\gamma = 0.08$, $SE = 0.04$, $p = 0.04$), thus supporting H3b. Simple slopes analyses indicated that this relationship was positive and significant at high levels (+1 SD; $\gamma = 0.17$, $SE = 0.06$, $p = 0.009$), but not significant at low levels (-1 SD) of turbulence ($\gamma = -0.03$, $SE = 0.08$, $p = 0.67$; $\Delta\gamma = 0.20$, $SE = 0.10$, $p = 0.04$). This interaction is illustrated in Fig. 3.

H1b, which posited that managers' perceived risk of being outcompeted would be positively related to their willingness to implement, was supported (Table 6; $\gamma = 0.23$, $SE = 0.08$, $p = 0.003$). Testing Hypothesis 4a, we also estimated the indirect effect of idea source on implementation willingness via the risk of being outcompeted at high (+1 SD) and low (-1 SD) levels of perceived turbulence. Consistent with our expectations, this relationship was stronger at high levels of technological turbulence ($\gamma = 0.04$, $SE = 0.02$, $p = 0.057$; 95 % CI [-0.001, 0.08]) but weaker at low levels of technological turbulence ($\gamma = -0.02$, $SE = 0.03$, $p = 0.36$; 95 % CI [-0.07, 0.03]).

Supporting H2b, managers' perceived risk of incompatibility was significantly negatively related to their willingness to implement the idea (Table 6, $\gamma = -0.35$, $SE = 0.08$, $p = 0.000$). Testing H4b, we also estimated the indirect effect of idea source on implementation willingness via risk of incompatibility at high (+1 SD) and low (-1 SD) levels of perceived turbulence. Supporting H4b, at high levels of technological turbulence, this relationship was stronger and significant ($\gamma = -0.06$, $SE = 0.03$, $p = 0.03$; 95 % CI [-0.11, -0.01]), while at low levels, it was not ($\gamma = 0.01$, $SE = 0.03$, $p = 0.67$; 95 % CI [-0.04, 0.06]). The direct, indirect, and total effects are included in Table 7.

Study 2 Discussion

In this quasi-experimental field study, we found that perceived

Table 5
Correlations and Descriptive Statistics (Study 2) with new idea filter.

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>Level 1 Variables</i>														
1. Idea Novelty	6.15	0.35	(0.98)											
2. Idea Usefulness	6.24	0.38	0.80**	(0.97)										
3. Idea Strategic Alignment	6.10	0.37	0.83**	0.76**	(0.95)									
4. Idea Source ^c	0.50	0.50	0.004	0.004	0.01									
5. Risk of Outcompetition	4.07	1.33	0.002	−0.004	−0.04	0.01	(0.95)							
6. Risk of Incompatibility	3.84	1.25	−0.04	−0.07*	−0.06 [†]	−0.01	0.45**	(0.95)						
7. Willingness to Implement	4.71	1.20	0.07*	0.09**	0.06*	−0.04	0.22**	−0.18**	(0.94)					
<i>Level 2 Variables</i>														
8. Manager Age	32.49	7.64	0.00	0.00	0.00	0.01	−0.01	0.06 [†]	−0.01					
9. Manager Gender ^a	0.23	0.48	0.00	0.00	0.00	−0.03	−0.14**	−0.07*	−0.09**	−0.05				
10. Manager Education ^b	0.97	0.16	0.00	0.00	0.00	0.02	−0.01	−0.16**	0.07*	−0.30**	−0.04			
11. Openness to Experience	4.62	0.98	0.00	0.00	0.00	0.01	0.07*	−0.03	0.12**	−0.27**	0.07*	0.14**	(0.77)	
12. Technological Turbulence	4.70	1.14	0.00	0.00	0.00	0.01	0.15**	0.05	0.17**	−0.003	−0.07*	0.03	0.12**	(0.88)

Notes: Variables 1 through 7 are idea-level (level 1) variables. Variables 8–13 are manager-level (level 2) variables. Idea-level correlations are shown below the diagonal and are based on idea scores ($N = 1025$). Manager-level correlations are shown above the diagonal and are based on manager scores ($N = 150$). Alpha reliabilities are presented along the diagonal.

^a Manager Gender (0 = Female; 1 = Male). ^b Education (0 = below college; 1 = college or above); ^c Idea source (1 = External, 0 = Internal).

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$.

Table 6
Unstandardized Results of Multilevel Path Model (Study 2).

	Risk of Outcompetition		Risk of Incompatibility		Willingness to Implement	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	4.05**	0.09	3.81**	0.08	4.75**	0.07
<i>Level 1 Predictors</i>						
Idea Novelty	0.42*	0.18	0.30*	0.14	−0.04	0.16
Idea Usefulness	0.07	0.13	−0.28*	0.11	0.23 [†]	0.13
Idea Strategic Alignment	−0.54**	0.15	−0.23 [†]	0.13	0.02	0.13
Idea Source ^c	0.07	0.05	0.04	0.06	−0.07	0.05
Risk of being Outcompeted					0.23**	0.08
Risk of Incompatibility					−0.35**	0.08
<i>Level 2 Predictors</i>						
Manager Age	0.002	0.01	0.003	0.01	0.01	0.01
Manager Gender ^a	−0.37*	0.17	−0.16	0.17	−0.24	0.15
Manager Education ^b	−0.29	0.61	−1.25*	0.60	0.44	0.56
Openness to Experience	0.12	0.09	0.02	0.09	0.16*	0.07
Technological Turbulence	0.04	0.08	0.00	0.08		
<i>Cross –Level Moderation</i>						
Idea Source X Technological Turbulence	0.11*	0.04	0.08*	0.05		
Pseudo- R^2	0.05		0.04		0.12	

Notes: Manager $N = 150$, Idea $n = 1025$. All Level 1 (idea-level) predictors have been group-mean centered, and Level 2 (manager) predictors have been grand-mean centered.

^a Manager Gender (0 = Female; 1 = Male). ^b Education (0 = below college; 1 = college or above).

^c Idea source (1 = External, 0 = Internal). [†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$.

technological turbulence interacted with the idea source to increase the risk of outcompetition, positively affecting managers' willingness to implement the idea. Additionally, perceived technological turbulence increased the risk of incompatibility of external sources, thereby negatively affecting managers' willingness to implement the idea. However, this study, while manipulating the idea source, did not manipulate technological turbulence. Therefore, in comprehensively establishing causality, we conducted another lab experiment to test our full model.

7. Study 3: Method

A power analysis conducted using G*Power, which assumed an alpha level of 0.05, a power of 0.8, and a small-to-medium effect size of 0.17 (Cohen, 1988)—adjusted downward to account for exploring interaction effects—indicated that we needed 382 participants for our final sample. In anticipation of high dropout rates common with online panels and potential issues such as incomplete responses or failure to pass attention checks (Meade & Craig, 2012), we recruited 442 U.S.

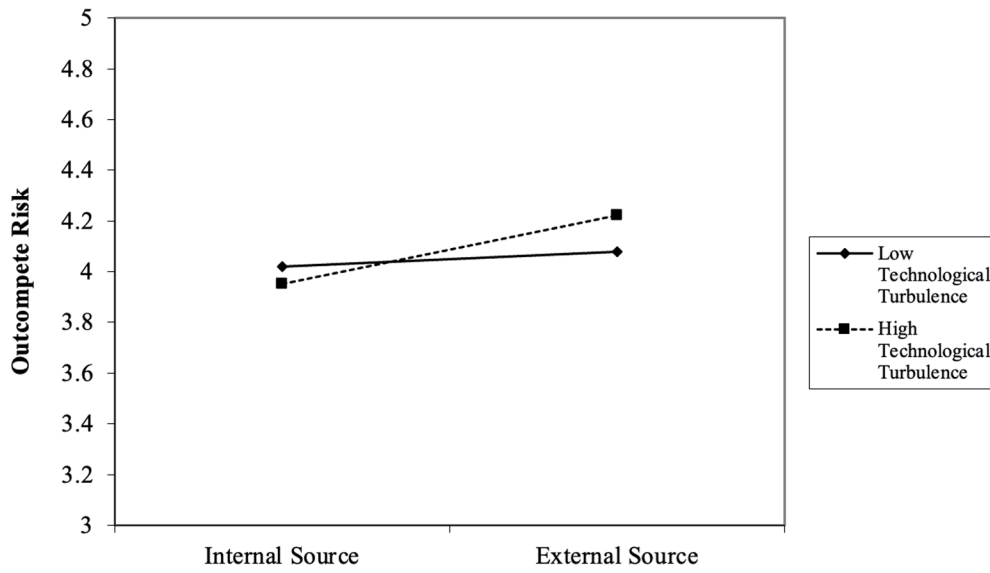


Fig. 2. Interaction of Idea Source and Technological Turbulence on Risk of Outcompetition (Study 2).

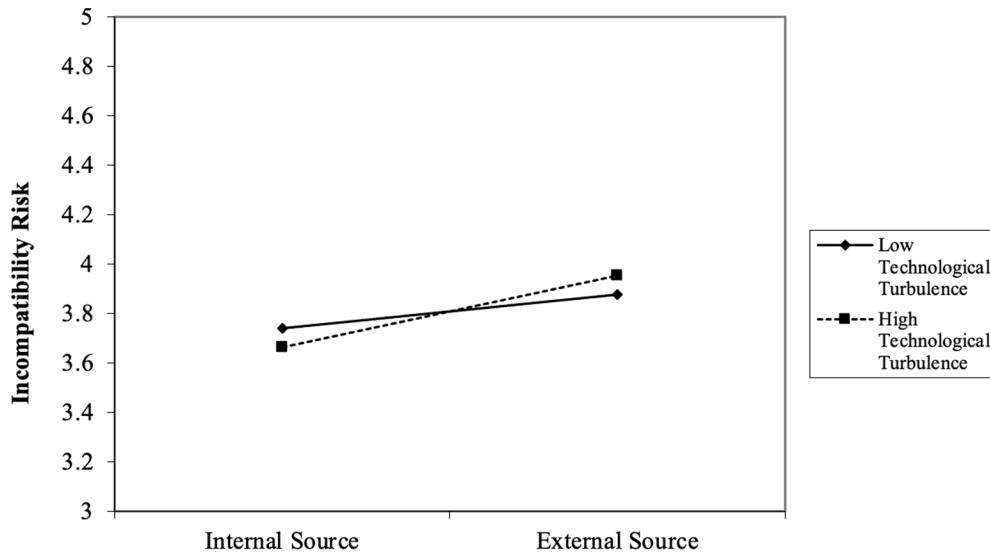


Fig. 3. Interaction of Idea Source and Technological Turbulence on Risk of Incompatibility (Study 2).

Table 7

Direct, Indirect, and Total Effects (Study 2).

Risk of Outcompetition					
Moderator	First Stage	Second Stage	Direct Effect	Indirect effect	Total Effect
High Tech Turbulence	0.17	0.23**	−0.07	0.04	−0.03
Low Tech Turbulence	−0.10	0.23**	−0.07	−0.02	−0.09
Risk of Incompatibility					
Moderator	First Stage	Second Stage	Direct Effect	Indirect effect	Total Effect
High Tech Turbulence	0.17**	−0.35**	−0.07	−0.06*	−0.13*
Low Tech Turbulence	−0.03	−0.35**	−0.07	0.01	−0.06

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

participants from Prolific to complete the study, offering a monetary compensation of \$1.50 each. This study was pre-registered (AsPredicted #161111). We recruited managers with at least 2 years of management experience. We excluded six participants who failed attention check questions (Meade & Craig, 2012), resulting in a retention of 436 participants. Of the final sample, 61.0 % were male, and 63.5 % were

Caucasian. The average age was 42.6 years ($SD = 11.6$), and the average work experience (part-time or full-time) was 19.6 years ($SD = 11.29$), with an average management experience of 9.47 years ($SD = 7.58$).

Employing a 2 (Idea source: external vs. internal) X 2 (Technological turbulence: high vs. low) between-subjects design, we randomly assigned participants to one of the four conditions. We used the same

experimental paradigm and creative idea used in Study 1.

Manipulations

Idea source manipulation. Participants in the external source condition ($N = 220$) read:

Taylor proposed that we consider the use of a new technology developed by an external research think tank. Taylor had read a recent report released by this external research institute detailing a new technology they had developed, which could be potentially applied to different situations. Taylor felt that this technology, which originates from an external source outside your organization, could be a viable solution capable of addressing your current challenges and capturing potential benefits.

Participants in the internal idea source condition ($N = 216$) read:

Taylor proposed that we consider the use of a new technology developed by your company. Taylor had read a recent report released by your company's research taskforce detailing a new technology that had been developed internally, which could be potentially applied to different situations.

Taylor felt that this technology, which originates internally within your organization, could be a viable solution capable of addressing your current challenges and capturing potential benefits.

Technological turbulence manipulation. We manipulated the characteristics of technological turbulence, which entails the emergence of new technologies in short development cycles and significant scale of technological changes (e.g., Chavez et al., 2015; Chen et al., 2018). Participants in the high technological turbulence condition ($N = 216$) read:

In your industry, the pace of technological innovation has been rapid. Changes in technology have been happening swiftly, with new technologies emerging in increasingly short development cycles, and the scale of these advancements is significant. These technological changes have provided opportunities for many new products and services in this industry. These rapid changes can quickly change market dynamics and competitive benchmarks, thus making them unpredictable.

Participants in the low technological turbulence ($N = 220$) read:

In your industry, the pace of technological innovation has been gradual and steady. Changes in technology have been happening at a gradual pace, with new technologies emerging in longer development cycles, and the scale of these advancements is small or moderate. Due to this stability, the opportunities for introducing new products and services in this industry have been limited. The chance of changes in market dynamics and competitive benchmarks has been low, thus making them predictable.

Measures.

Participants rated the following measures using a 7-point Likert-type scale (1 = *Strongly Disagree* to 7 = *Strongly Agree*).

Risk of Outcompetition. Participants rated this using the same three items from previous studies ($\alpha = 0.83$).

Risk of Incompatibility. Participants used the same three items as in previous studies to rate the idea's risk of incompatibility with existing practices ($\alpha = 0.81$).

Willingness to Implement. Participants responded to the same three items from previous studies and measured their willingness to implement the proposed idea ($\alpha = 0.94$).

8. Study 3: Results And Discussion

8.1. Manipulation checks

To assess the effectiveness of the idea source manipulation, we used the same items from Study 1. For the technological turbulence manipulation, participants assessed the three items from Jaworski and Kohli's (1993) measure (e.g., "The technology in our industry is changing rapidly," $\alpha = 0.93$). In testing the effectiveness of the idea source and the technological turbulence manipulations, results demonstrated that the main effects of idea source on the manipulation check items for the

internal and external idea sources were significant, but did not affect the manipulation checks for turbulence. Specifically, participants in the external idea source condition ($M = 6.00$, $SD = 1.32$) were more likely to identify the idea source as external compared to those in the internal source condition ($M = 2.59$, $SD = 1.88$), $F(1,432) = 487.31$, $p < 0.001$, $\eta_p^2 = .53$. Similarly, participants in the internal idea source condition ($M = 5.94$, $SD = 1.29$) were more likely to perceive the idea as internally sourced compared to those in the external condition ($M = 2.29$, $SD = 1.62$), $F(1,434) = 692.97$, $p < 0.001$, $\eta_p^2 = 0.62$). For the technological turbulence manipulation, participants in the high technological turbulence condition rated the manipulation check items higher ($M = 6.08$, $SD = 0.92$) than those in the low turbulence condition ($M = 4.53$, $SD = 1.68$), $F(1,432) = 141.14$, $p < 0.001$, $\eta_p^2 = 0.25$). This manipulation did not affect the manipulation check items of the source conditions. These results confirmed the effectiveness of both of our manipulations.

8.2. Hypothesis testing

Descriptive statistics and correlations among core variables are presented in Table 8, and the results of our regression analyses are presented in Table 9. There was a significant main effect of idea source on participants' willingness to implement the idea, such that they were less likely to implement external ($M_{\text{external}} = 5.31$, $SD = 1.23$) as compared to internal ideas ($M_{\text{internal}} = 5.61$, $SD = 1.09$; $\Delta M = -0.30$, $F(1,434) = 7.15$, $p = 0.008$, $\eta_p^2 = 0.02$). One-way ANOVA results revealed that external ideas ($M_{\text{external}} = 4.98$, $SD = 1.22$) were perceived as significantly riskier in terms of being outcompeted than internal ones ($M_{\text{internal}} = 4.15$, $SD = 1.40$; $\Delta M = 0.84$, $F(1,434) = 44.47$, $p < 0.001$, $\eta_p^2 = 0.09$), supporting H1a. Similarly, external ideas ($M_{\text{external}} = 4.24$, $SD = 1.23$) were perceived as having a significantly higher risk of incompatibility than internal ones ($M_{\text{internal}} = 3.48$, $SD = 1.42$; $\Delta M = 0.76$, $F(1,434) = 36.12$, $p < 0.001$, $\eta_p^2 = 0.08$), supporting H2a.

In testing H3a, a two-way ANOVA with idea source and technological turbulence as factors and the perceived risk of outcompetition as the DV revealed a significant two-way interaction ($F(1,432) = 6.98$, $p = 0.009$, $\eta_p^2 = 0.02$; see Fig. 4). To explore this interaction further, we compared the marginal means across conditions (see Table 10 for estimated marginal means). Results showed that within the high technological turbulence conditions, external ideas were rated as significantly riskier ($M_{\text{external, high tech turb}} = 5.27$, $SD = 1.04$) compared to internal ideas ($M_{\text{internal, high tech turb}} = 4.10$, $SD = 1.48$; contrast = 1.17, $SE = 0.18$, $p < 0.001$). Within the low technological turbulence conditions, external ideas ($M_{\text{external, low tech turb}} = 4.69$, $SD = 1.31$) were still rated as significantly riskier than internal ideas ($M_{\text{internal, low tech turb}} = 4.19$, $SD = 1.33$), although the difference was smaller in magnitude (contrast = 0.51, $SE = 0.18$, $p = 0.004$).

In testing H3b, two-way ANOVA with idea source and technological turbulence as factors and the perceived risk of incompatibility as the DV showed the two-way interaction, while in the expected pattern, was only marginally significant ($F(1, 432) = 3.10$, $p = 0.079$, $\eta_p^2 = 0.007$; see Table 11 for estimated marginal means). Thus, H3b was not supported.

Supporting H1b, the regression of willingness to implement on the perceived risk of outcompetition, controlling for the risk of incompatibility, showed a significant positive relationship ($B = 0.19$, $p < 0.001$). Testing the mediation model implied by H1a and H1b using the PROCESS macro (model 4; 5000 bootstrapping iterations; Hayes, 2013) showed that the indirect effect of external idea source on implementation willingness via managers' perceived risk of outcompetition was significant (estimate = 0.17, $SE = 0.04$, 95 % CI [.09, 0.26]).⁵ Further, we used the PROCESS macro (model 7) to test H4a, estimating the

⁵ Similar to Study 1, we also explored alternative mechanisms, such as misalignment with organizational values, cost, risk of idea theft, and group defensiveness. This did not change our results in any way. Results of these analyses are provided in Online Appendix 3.

Table 8
Correlations and Descriptive Statistics (Study 3).

	<i>M</i>	<i>SD</i>	1	2	3	4	5
1. Idea Source ^a	0.50	0.50					
2. Technological Turbulence ^b	0.50	0.50	0.02				
3. Risk of being Outcompeted	4.57	1.38	0.31**	0.10*	(0.81)		
4. Risk of Incompatibility	3.86	1.38	0.28**	−0.04	0.24**	(0.83)	
5. Willingness to Implement	5.46	1.17	−0.13**	0.07	0.10*	−0.48**	(0.94)

Notes: *N* = 436. Alpha reliabilities are presented along the diagonal.

^a Idea source (1 = External, 0 = Internal). ^b Technological Turbulence (1 = High, 0 = Low).

* *p* < 0.05; ***p* < 0.01.

Table 9
Regression Analysis of Risk of Outcompetition, Risk of Incompatibility, and Willingness to Implement (Study 3).

	Risk of Outcompetition		Risk of Outcompetition		Risk of Incompatibility		Risk of Incompatibility		Willingness to Implement		Willingness to Implement	
	<i>b</i>	<i>SE</i>	<i>b</i>	<i>SE</i>	<i>b</i>	<i>SE</i>	<i>b</i>	<i>SE</i>	<i>b</i>	<i>SE</i>	<i>b</i>	<i>SE</i>
Idea Source ^a	0.84***	0.13	0.51**	0.18	0.76***	0.13	0.54**	0.18	−0.30**	0.11	−0.13	0.10
Tech Turbulence ^b			−0.09	0.18			−0.34 [†]	0.18			0.07	0.10
Idea Source × Tech Turbulence			0.66**	0.25			0.45 [†]	0.25				
Outcompetition Risk											0.20***	0.04
Incompatibility Risk											−0.44***	0.04
<i>R</i> ²	0.09		0.12		0.08		0.09		0.02		0.28	

Notes: *n* = 436; table entries represent unstandardized parameter estimates with standard errors.

^a Idea Source (External = 1, Internal = 0); ^b Technological Turbulence (High = 1, Low = 0).

[†]*p* < 0.10; **p* < 0.05; ***p* < 0.01; ****p* < 0.001

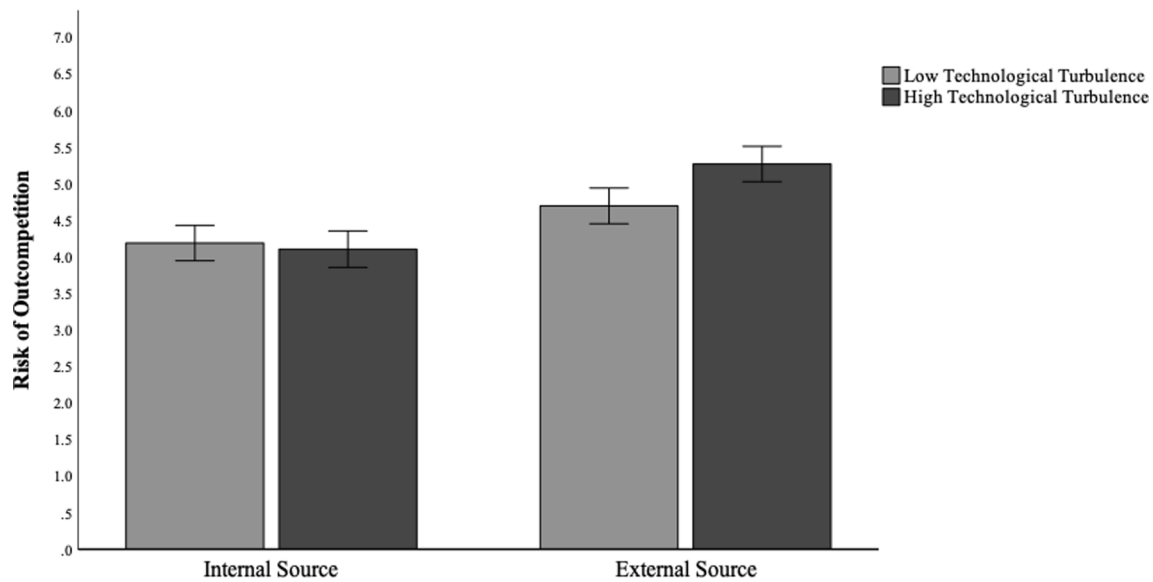


Fig. 4. The Interactive Effect of Idea Source and Technological Turbulence on the Risk of Outcompetition (Study 3).

indirect effect of idea source on implementation willingness via the risk of outcompetition at high (+1 *SD*) and low (−1 *SD*) levels of turbulence. Consistent with our expectations, the indirect effect was stronger at high levels of technological turbulence (estimate = 0.24, *SE* = 0.06; 95 % CI [.13, 0.36]) compared to low levels (estimate = 0.10, *SE* = 0.04, 95 % CI [.03, 0.19]; Δ estimate = 0.13, *SE* = 0.06, 95 % CI [.03, 0.27]), supporting H4a. The direct, indirect, and total effects are included in Table 12.

Supporting H2b, the regression of willingness to implement on the perceived risk of incompatibility, controlling for the risk of outcompetition, revealed a significant negative effect (*B* = −0.45, *p* < 0.001). Testing the mediation model (implied by H2a-b) using the

PROCESS macro (model 4), the indirect effect of external idea on implementation willingness via perceived risk of incompatibility was significant (estimate = −0.34, *SE* = 0.07, 95 % CI [−0.47, −0.21]). Given the marginal interaction effect (H3b), we did not test the conditional indirect effects (H4b).

9. General discussion

The current research examines how managers make decisions about implementing creative ideas originating from sources external to their organization. Across multiple studies, we find that managers perceived two key risks. First, external ideas were perceived to carry higher risks of

Table 10

Estimated Marginal Means of Risk of Outcompetition across All Experimental Conditions (Study 3).

Risk of Outcompetition	Technological Turbulence					
	High Turbulence			Low Turbulence		
	Mean	S.E.	N	Mean	S.E.	N
External Source Idea Condition	5.27 ^a	1.04	111	4.69 ^{b, c}	1.31	109
Internal Source Idea Condition	4.10 ^{b, d}	1.48	105	4.19 ^{b, d}	1.33	111

Note: Based on Bonferroni comparisons, the value in the “a” cell is significantly higher than the value in the “b” cell.
The value in the “c” cell is significantly higher than the value in the “d” cell. Other sets of contrasts are not significant.

Table 11

Estimated Marginal Means of Risk of Incompatibility across All Experimental Conditions (Study 3).

Risk of Incompatibility	Technological Turbulence					
	High Turbulence			Low Turbulence		
	Mean	S.E.	N	Mean	S.E.	N
External Source Idea Condition	4.29 ^a	1.24	111	4.18 ^c	1.22	109
Internal Source Idea Condition	3.30 ^b	1.37	105	3.64 ^b	1.45	111

Note: Based on Bonferroni comparisons, the value in the “a” cell is significantly higher than the value in the “b” cell.
The value in the “c” cell is significantly higher than the “b” cell. Other sets of contrasts are not significant.

Table 12

Direct, Indirect, and Total Effects (Study 3).

Risk of Outcompetition					
Moderator	First Stage	Second Stage	Direct Effect	Indirect effect	Total Effect
High Tech Turbulence	1.17***	0.20***	−0.13	0.24***	0.11**
Low Tech Turbulence	0.51**	0.20***	−0.13	0.10*	−0.03
Risk of Incompatibility					
Moderator	First Stage	Second Stage	Direct Effect	Indirect effect	Total Effect
High Tech Turbulence	0.99***	−0.44***	−0.13	−0.44***	−0.31**
Low Tech Turbulence	0.54***	−0.44***	−0.13	−0.24**	−0.11**

** $p < 0.01$; *** $p < 0.001$.

outcompetition, which positively influenced their implementation willingness. Second, external ideas were viewed as being more incompatible with existing organizational processes, which reduced implementation willingness. Further, managers’ perceptions of technological turbulence amplified both the perceived risks associated with external ideas. Thus, in turbulent environments, managers faced heightened pressure to adopt external ideas to stay competitive, yet simultaneously experienced greater uncertainty about successfully integrating these ideas into existing systems. The current study makes some important theoretical contributions to the managerial creative idea endorsement and innovation management literatures.

9.1. Theoretical contributions

First, we integrate insights from the psychological literature on managerial creativity endorsement with the macro literature on innovation management (e.g., DCT; Teece et al., 1997) to shed nuanced light on the dynamics of external idea implementation. While prior work on managerial endorsement finds that managers often reject creative ideas (e.g., Mueller et al., 2012), it has been agnostic to the source of these

ideas—a significant oversight, given that external ideas catalyze innovation (Laursen & Salter, 2006). In this regard, research on the Not-Invented-Here (NIH) syndrome documents that organizations often reject knowledge originating outside the organizational boundary (Antons & Piller, 2015; Agrawal et al., 2010) yet cannot explain findings in the broader innovation management domain that suggests that decision-makers are also receptive to external knowledge (Afuah & Tucci, 2012; Piezunka & Dahlander, 2015). By applying DCT to the managerial creative idea endorsement literature, we identify a fundamental tradeoff managers face: balancing the need to seize competitive opportunities against maintaining internal capabilities without disruption (Teece et al., 1997). Understanding this tension is crucial as it helps organizations adapt effectively while mitigating risks. Organizations resisting external ideas risk obsolescence, whereas those adopting too aggressively risk internal failure. This perspective enables informed decision-making by helping managers assess trade-offs, reduce costly mistakes, and determine the right timing for innovation.

Second, and related to the above, we unpack the mechanisms underlying managers’ implementation of external ideas. While the idea of risk is not new to the managerial creativity endorsement literature, it has tended to characterize risk uniformly as an impediment to idea adoption (Johnson & Proudfoot, 2024; Lane et al., 2022). Our findings reveal a more complex relationship: while incompatibility risks arising from committing to implement an external idea can deter managers from adopting external ideas, we find that potential competitive risks that can emerge from omitting to adopt these ideas can encourage their adoption. Thus, our dual-risk framework enriches the managerial creativity endorsement literature by shifting the focus from risk as a uniform barrier to highlighting that certain risks can motivate managers to endorse external ideas, while others can inhibit idea implementation. We demonstrate that the risk of being outcompeted potentially drives implementation, while the risk of system incompatibility constrains organizational change. By systematically controlling for several alternative explanations identified in related literatures (e.g., NIH; Antons & Piller, 2015), we show that perceived risks of outcompetition and system incompatibility are distinct psychological mechanisms driving managers’ willingness to implement external creative ideas.

Although we find support for a dual-risk pathway, a meta-analysis across all our studies (McShane & Böckenholt, 2017) indicates a significant negative overall effect of external idea source on managerial implementation willingness (see Online Appendix 9). This suggests that the risk of incompatibility may potentially play a stronger role in influencing such decisions. This overall bias toward internal ideas likely reflects managers’ tendency to avoid the risk of commission (adopting an external idea that might cause incompatibility) in favor of the seemingly safer option of internal ideas. This aligns with the concept of omission bias, where people prefer inaction over action. For example, research by Ritov and Baron (1990) showed that people judge harmful actions more negatively than harmful inactions, even when both result in identical outcomes with the same severity and probability. Relatedly, Spranca et al., (1991) demonstrated that people often avoid action to escape responsibility. In the context of external ideas, this bias reinforces the perception that action typically generates more regret than inaction, even with equally harmful outcomes (Zeelenberg et al. 1998).

Additionally, while we found support for our first-stage moderation hypotheses (H3a, H3b), where technological turbulence moderates the relationship between idea source and risk perceptions, our analyses showed that the direct effect of the interaction on the DV was not statistically significant in both Study 2 ($b = 0.05$, $SE = 0.04$, $p = 0.27$) and Study 3 ($b = 0.16$, $SE = 0.22$, $p = 0.48$). These non-significant interaction effects directly on the implementation decisions suggest that the opposing mediating mechanisms may have potentially canceled each other out under high turbulence. Therefore, compared to examining the overall relationship, our dual risk pathway model provides a more complete understanding of how managers react to external ideas during turbulence. This highlights the value of considering multiple, even

opposing, mechanisms when studying innovation decisions under uncertainty and opens up the field to consider what factors create conditions under which one risk outweighs the other.

Third, while prior research on managerial creativity endorsement has primarily examined the characteristics of the idea, the proposer, the evaluator, or the organizational context (e.g., Lu et al., 2019; Mueller et al., 2018; Zhou et al., 2019), our study shifts the focus to a key environmental factor—technological turbulence. We highlight how managers' perceptions of this turbulence serve as a critical boundary condition, shaping their risk assessments and willingness to implement external ideas. We find that high perceived technological turbulence simultaneously heightens both competitive and incompatibility risk perceptions, which can potentially lead to decision paralysis precisely when timely adaptation is most critical. This aligns with prior research showing that when decision-makers face equally strong opposing considerations and trade-offs between valued attributes, they experience selection difficulty, which can result in choice deferral, status quo bias, or omission bias (Anderson, 2003). By highlighting this paradox—i.e., when organizations most need external ideas to stay competitive, managers may be most inclined to perceive these ideas as threats to existing capabilities—our work extends DCT by revealing tensions that the theory may not have fully addressed.

Our work also extends the DCT literature in other important ways. While DCT articulates what organizational capabilities are needed for competitive advantage in dynamic environments, it remains largely silent on the psychological mechanisms through which these organizational imperatives manifest in managerial decision-making. DCT assumes harmony among organizational capabilities without considering how managers might navigate competing considerations. While Teece (2007) introduced the concept of “microfoundations” as building blocks of dynamic capabilities, this primarily addresses organizational processes rather than psychological factors. By identifying managers' perceptions of risks of incompatibility versus competitive urgency, as well as their perceptions about the technological environment as important predictors of their willingness to implement creative ideas, our work demonstrates that operational decisions about competitive dynamics and dynamic capabilities are heavily influenced by individual managerial psychology. This suggests that the study of how organizations develop capabilities and innovative advantages would be incomplete without studying how managers' psychological perceptions of the situation may impact their decision-making in leveraging various external opportunities.

Relatedly, the dynamic capabilities framework often assumes that managers can rationally assess and reconfigure resources. However, studying competing managerial considerations in external idea evaluation highlights the cognitive biases and risk aversion that influence managerial decisions, enriching the behavioral aspects of the framework. Our research suggests that in the context of evaluating external ideas for implementation, although they consider competing risks, managers are predominantly concerned with minimizing incompatibility with existing capabilities, thereby foregoing the opportunity to learn from external ideas and thus adapt their internal capabilities.

Our integration with DCT also provides new insights into the managerial creativity endorsement literature, which has identified middle managers as critical bottlenecks in the innovation process (Mueller et al., 2012; Lu et al., 2019). While this work has focused on managers' individual differences and other contextual factors, including organizational culture, hierarchy, and field-specific practices (Keum & See, 2017; Mount et al., 2021; Mueller et al., 2018; Zhou et al., 2017), we provide new insights into potential reasons for the “innovation gap.” Specifically, our research goes beyond existing explanations by demonstrating how strategic and practical risk-related considerations and the perception of environmental contexts shape managers' decision-making. By revealing the complex psychological mechanisms that drive managers' reluctance to implement external ideas, we demonstrate that

creativity endorsement is fundamentally a sophisticated risk negotiation process deeply embedded in strategic considerations and dynamically responsive to environmental contexts.

9.2. Practical Implications

Our research offers valuable implications for effective innovation decision-making, especially in periods of technological turbulence. Given the critical role managers play in evaluating and endorsing ideas, organizations should develop targeted tools that help managers simultaneously assess both the competitive potential and systemic integration challenges of external ideas. Structured evaluation frameworks should explicitly guide managers through assessing two key risk dimensions: first, the potential competitive advantages of an external idea, and second, its potential disruption to existing organizational systems. Such frameworks can help managers choose the right time to implement external innovations and phase implementation instead of adopting reactive decision-making and disruptive changes all at once.

During periods of technological turbulence, organizations can help managers constructively manage competing risk considerations by developing sophisticated scenario-planning tools that comprehensively evaluate external innovations. Such tools should enable managers to anticipate both how external innovations might shape the competitive landscape and the potential challenges of integrating these ideas into existing organizational processes. Organizations would benefit from developing early-access partnerships with external innovators to facilitate this process. These strategies can enable managers to better understand how external innovations may align with evolving market demands while assessing compatibility issues.

9.3. Strengths, Limitations, and Future Directions

This study employs a multi-method approach, integrating qualitative, field, and experimental designs to test our hypotheses. Real managers participated in all our studies, providing ecological validity for our results. In Study 2, we used a quasi-experimental design with employee-generated ideas and manipulated idea sources, establishing causality while also maintaining contextual relevance. By triangulating evidence from different methodologies and samples, we provide robust support for our hypotheses.

Despite these strengths, our research has some distinct limitations that point to areas for future research. First, while we defined external ideas based on organizational boundaries, scholars also suggest other ways to conceptualize externality through industry, geographic, and knowledge context boundaries (e.g., Antons & Piller, 2015; Shannon & Weaver, 1964). In a supplementary study (Study A; see Online Appendix 6) comparing internal ideas to external ideas from both the same and different industries, we found that both external conditions evoked higher perceived risks than internal ideas, with no significant difference between the two external conditions. This suggests that organizational boundaries play a crucial role in driving managers' reactions. However, investigating other externality gradients would be a productive avenue for future work. Second, while holding idea characteristics constant across studies helped isolate source effects, this approach limited our understanding of how managers evaluate different idea types across varied contexts. Future research could explore how our findings might vary with ideas of different complexity, from varying domains, etc., which would better reflect real-world innovation scenarios and potentially reveal insights that our controlled design couldn't capture.

Third, while our findings support incompatibility and out-competition risks as key drivers of idea source effects due to their theoretical prominence and empirical strength in our studies, other mechanisms, such as cost concerns and values alignment, may become more influential in specific contexts. Our research also doesn't fully explore licensing and open innovation strategies, which present unique trade-offs—licensing might reduce incompatibility risks but limit

adaptation flexibility, while open innovation might facilitate customization but introduce potential goal misalignment. Fourth, our measure of implementation willingness may not perfectly translate to actual implementation behavior. Future research could employ longitudinal studies tracking real implementation processes to better capture the organizational dynamics mediating innovation adoption across these varied contexts. The interplay between these various mechanisms likely creates complex decision environments that managers must navigate when evaluating external ideas. Understanding these nuanced interactions could provide valuable insights for organizations seeking to optimize their innovation sourcing strategies while minimizing implementation barriers.

Our research on understanding competing mechanisms also opens up opportunities for the field to consider the factors determining when one risk outweighs the other. Future studies should consider how idea characteristics, individual differences in risk-taking tendencies, and organizational factors like competitive intensity may influence whether managers prioritize outcompetition or incompatibility risks (Sitkin & Pablo, 1992; Auh & Menguc, 2005). Additionally, while our current model focuses primarily on negatively framed risks, extending the framework to consider the positive potential of external ideas—such as gaining a competitive advantage or radically improving processes—could add valuable nuance. Future research could also explore how decision-makers resolve these tensions in practice, potentially through sequential attention to different risks, the development of organizational routines that address both risks simultaneously, and the development of ambidextrous capabilities that allow organizations to manage the paradox rather than resolve them (Smith & Tushman, 2005).

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CRediT authorship contribution statement

Rebekah SungEun Hong: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Vijaya Venkataramani:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mengxi Yang:** Validation, Resources, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.obhdp.2025.104417>.

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